

TESLA OPTIMUS

General Purpose Bi-Pedal Humanoid Robot

Introduction

Announced in 2021, the Tesla Optimus (also known as Tesla Bot) represents Tesla's ambitious entry into the world of general-purpose robotics. Leveraging the artificial intelligence and computer vision technologies developed for Tesla's Autopilot and Full Self-Driving (FSD) systems, Optimus is designed to navigate human environments and perform tasks that are dangerous, repetitive, or boring.

"The goal is to build a general-purpose, bi-pedal, humanoid robot capable of performing tasks that are unsafe, repetitive or boring." — Tesla

Design Philosophy and AI Integration

Optimus is unique because it is designed for mass production from the ground up. Tesla aims to manufacture millions of units, driving the cost down significantly. The robot uses the same neural networks and camera-based vision systems found in Tesla vehicles to perceive its environment.

Expected Specifications (Gen 2)

Height	~5 ft 8 in (173 cm)
Weight	~121 lbs (55 kg)
Carrying Capacity	45 lbs (20 kg)
Actuators	Tesla-designed custom actuators